

ORIGINAL RESEARCH ARTICLE

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Soybean profitability and yield component response to nitrogen fertilizer in Iowa

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Abstract

Nitrogen fertilizer application to soybean [*Glycine max* (L.) Merr.] in Iowa, USA, has shown inconsistent results. We performed a study in central Iowa (2015 and 2016) to investigate the effect of N fertilizer rate (0, 45, 90, 135 kg N ha⁻¹) and application timing (planting, flowering, pod setting) on soybean yield, yield components, and to calculate the economic net return to N fertilizer. Results showed a positive effect of N fertilizer on soybean yield and yield components both years. Seed and aboveground biomass dry weight were positively correlated to N fertilizer, and both were 17% greater than No-N treatment. Nitrogen fertilizer rate that significantly increased seed and aboveground biomass was 135 kg N ha⁻¹ regardless of application timing (2015), or at planting (2016). Moreover, the same N fertilizer addition applied at planting benefitted seed and aboveground biomass N accumulation only in 2016 (avg. 32.00 and 34.68 g N uptake m⁻², respectively), both 1.5-times higher than No-N treatment. Favorable environmental conditions during 2016 lead to hand-measured yield difference of 22% compared to 2015. Economic net return analysis showed that the additional revenue from increased yield attributed to supplemental N fertilization offset the application cost, resulting in net return gains between US\$5.83 to \$281.89 ha⁻¹ (all treatments except 45 kg N ha⁻¹ on 2015). This study highlights the importance to parse out soybean yield in its components, and the need to quantify yield gains from N fertilizer additions in economic terms which shed some light on any tradeoffs.

1 | INTRODUCTION

Soybean [*Glycine max* (L.) Merr.] grain production (100 million Mg in 2017) in the midwestern United States contributes approximately US\$20 billion yr⁻¹ to the U.S. economy (USDA-NASS, 2018). Historical data from the

midwestern United States show an average yield gain of 20 kg ha⁻¹ yr⁻¹ from 1927 to 2014, with noticeable variation in production due to annual weather variability (Egli & Hatfield, 2014). The historical yield gain is attributed to genetic advances (e.g., De Bruin & Pedersen, 2008; Koester, Skoneczka, Cary, Diers, & Ainsworth, 2014; Kumudini, Hume, & Chu, 2001) and management improvements (van Roekel & Purcell, 2014; Marburger, Mourtzinis, Gaska & Conley, 2016; Sindelar, Schmer, Jin, Wienhold, & Varvel,

Abbreviations: BNF, biological nitrogen fixation; DOY, day of the year; NHI, nitrogen harvest index; TC, total carbon; TN, total nitrogen.

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2016). For instance, in Iowa, after the commercial release of the first transgenic soybean seed in 1996 (Fernandez-Cornejo & McBride, 2002) soybean yields increased at a rate of $38 \text{ kg ha}^{-1} \text{ yr}^{-1}$ equal to a 30% of annual yield boost between 1996 and 2018 (USDA-NASS, 2018).

Nitrogen fertilization has contributed substantially to historical maize (*Zea mays* L.) yield increases. However, soybean, as a leguminous crop, relies on obtaining N from two sources: biological N fixation (BNF) and from the soil mineral N pool (Unkovich et al., 2008). The contribution of BNF to total N uptake is variable from year-to-year, among cultivars, locations, and management practices, and typically ranges from 20 to 80% (Harper, 1974; Salvagiotti et al., 2008). According to the literature, BNF is suppressed by high levels of endogenous soil mineral N (Schipanski, Drinkwater, & Russelle, 2010) and by supplemental N fertilizer additions (e.g., Gelfand & Robertson, 2015; Goss, de Varennes, Smith, & Ferguson, 2002; Harper & Hageman, 1972). In some cases, N applications at the full flowering stage could reduce BNF up to 16% (Tamagno, Sadras, Haegerle, Armstrong, & Ciampitti, 2018). Many soybean N fertilizer trials highlighted that N-limitation exist in soybean fields while cultivated under standard practices (particularly no-N fertilizer addition) in high-yielding environments. Thus, the aim to mitigate this N-limitation and reinforced BNF is by the addition of N fertilizer, in some cases soon after planting to avoid affecting nodule formation (Abendroth, Elmore, & Ferguson, 2006; Ambrosini et al., 2019), others guided by additions at different soybean developmental stages (Haq & Mallarino, 2000, 2005; Schmitt, Lamb, Randall, Orf & Rehm, 2001; Cafaro La Menza, Monzon, Specht, & Grassini, 2017, 2019).

A large body of literature suggests that high-yielding soybean systems can be N limited, in which BNF or soil mineral N pool cannot keep up with the high N demand (Ciampitti & Salvagiotti, 2018; Rotundo, Borrás, De Bruin, & Pedersen, 2014; Tamagno et al., 2018). According to Santachiara, Borrás, Salvagiotti, Gerde, and Rotundo (2017), high soybean seed yield and N accumulation do not necessarily correspond to a higher contribution of BNF. If we assume a soybean seed N concentration of 5.8% (Tamagno et al., 2017), and 0.44 kg N is biologically fixed $\text{ha}^{-1} \text{ kg}^{-1}$ N accumulated ha^{-1} (Ciampitti & Salvagiotti, 2018; Córdoba et al., 2019), a soybean yield increase from 3.5 Mg ha^{-1} (i.e., state average, USDA-NASS, 2018) to 5.0 Mg ha^{-1} will require an additional 87 kg N Mg^{-1} seeds, or 119 kg N ha^{-1} increase in total biomass uptake if N harvest index (NHI) is equal to 0.73 (Salvagiotti et al., 2008). Theoretically, this represents an increase of 43% in BNF. However, soybean BNF is an energetically expensive process that requires approximately 7.06 g of glucose to biologically fix 1 g of N (Boote, Hoogenboom, Ingram, & Jones, 2008); thus a 43% increase in BNF will represent a high energetic cost to the

Core Ideas

- Nitrogen fertilizer, along with favorable conditions, had a positive effect on soybean yield components.
- Hand-harvested soybean yield increases of 0.03 and 0.10 kg m^{-2} were realized from N fertilization.
- Nitrogen fertilization addition results in positive net returns between US\$5.33 and $\$281.89 \text{ ha}^{-1}$.

plant (Herridge & Rose, 2000; Silsbury, 1977). However, the energetic cost of BNF in soybean remains controversial, some studies have found similar or contrasting results (Andrews, Lea, Raven, & Azevedo, 2009; Santachiara et al., 2017; Tamagno et al., 2018), and others have investigated the role of supplemental N fertilization that can increase soybean productivity in high-yielding fields to meet this high N demand (Cafaro La Menza et al., 2017, 2019; Ortéz, Tamagno, Salvagiotti, Prasad, & Ciampitti, 2019).

Our literature review of 22 studies on supplemental N fertilization to soybean in the Midwest (Table 1) showed that 64% of the studies found a significant yield response to N fertilizer from which almost half of them applied before or at planting (e.g., Bender, Haegerle, & Below, 2015; Osborne & Riedell, 2006; Sorensen & Penas, 1978) and the rest applied at determinate phenological stages between V5 to R6.5 (e.g., Garcia & Hanway, 1976; Haq & Mallarino, 2005; Ortéz et al., 2019). Moreover, nearly all the rainfed studies failed to report an economic benefit to N fertilization, with the only two exceptions being Wesley, Martin, and Duncan (1999) and Orlowski et al. (2016). Yield increase is relevant for farmers, but not the most important factor to add N fertilization. Net revenue increase is the essential motivator for farmers to apply the practice. It is critical to report an economic analysis from high-input management system studies with significant responses from N fertilizer applications on soybean fields. Accounting for the cost of N fertilization, yield response, and commodity price is a critical decision process for farmers and agronomists to make the correct management decisions for a given year. Any gain or loss in soybean production by the addition of N fertilizer should be reported in economic terms. Unfortunately, many high-input soybean management studies fail to show break-even possibilities in economic returns to input investment (Supplemental Table S1). Some recent studies found that economic returns can benefit by increasing soybean seed price, and boost soybean yield by the increasing use of agrochemicals and management intensity (Greer, Martins, While, & Pittelkow, 2020; Orlowski et al., 2016).

TABLE 1 Management practices showing locations, activities, and dates during 2015 and 2016

Activities	2015		2016	
	Date	DOY	Date	DOY
Nitrogen fertilizer application at:				
Planting	13 May	132	19 May	139
Stage R1/R2	26 June	176	30 June	181
Stage R3/R4	15 July	195	31 July	212
Herbicide application	20 June	170	18 July	199
Hand harvest	30 Sept.	272	20 Sept.	263
Mechanical harvest	26 Oct.	298	11 Oct.	284

Note. DOY, day of the year.

Another interesting result from our review was the absence of N fertilization experiments in Iowa over the last decade (Ruiz Diaz, Pedersen, & Sawyer, 2009; Table 1). Studies on rainfed environments within the Midwest, the United States reported an average soybean yield of 3.09 Mg ha⁻¹ (range 1.22–4.95 Mg ha⁻¹; Table 1), whereas studies with irrigation treatments reported an average yield of 4.32 Mg ha⁻¹ (range 1.93–6.70 Mg ha⁻¹; Table 1). Cafaro La Menza et al. (2017) and Cafaro La Menza et al., 2019 reported maximum soybean yield levels in Nebraska could possibly reach 6.70 Mg ha⁻¹, but not easily achievable with current management, and with an excessive application of 870 kg N ha⁻¹. On the other hand, soybean yield gains and economic returns obtained in high-input managed fields in the U.S. Midwest are site-specific (Gaspar, Marburger, Mourtzinis, & Conley, 2014; Rattalino Edreira et al., 2017) and may not offset the negative environmental performance in terms of energy, N leaching potential, partial global warming potential, and agrochemical contamination risks (Greer et al., 2020). An extensive study conducted by Mourtzinis et al. (2018) with a combined dataset that included 207 environments from soybean N fertilizer trials in the United States, found that 68% of the yield variability was explained by environmental conditions (e.g., growing season weather) and to a lesser extent to N fertilizer application.

Looking into previous N fertilizer trails conducted in Iowa from 1971 to 2019, we found six studies with confounding results. Four of the six studies showed a positive effect of N fertilization on soybean production (Garcia & Hanway, 1976; Hanway & Weber, 1971; Haq & Mallarino, 2000, 2005) and two studies with a negligible effect (Barker & Sawyer, 2005; Ruiz Diaz et al., 2009). Although soybean genetic improvements aim to boost yield, crop growth rates have increased over time from ranges of 90–110 to 138–213 kg ha⁻¹ d⁻¹ during the 1970s to 2015, respectively (Córdova et al., 2019; Hanway & Weber, 1971). The increase of crop growth may have also increased of nutrient uptake partitioning to different plant organs. One recent study by De Bruin and Pedersen (2008) tested

new vs. old soybean cultivars in three different locations in Iowa and found that indeed new soybean cultivars had 18% higher yields compared with old soybean cultivars. According to Ortéz et al. (2018) in the United States newer soybean cultivars released in the 2010s have increased soybean yields by 29% compared with old cultivars from the 1980s (on average 2,909 to 4,073 kg ha⁻¹). This might have been due to the breeding success in which modern midwestern soybean has increased light interception efficiency of photosynthetically active radiation up to 90%, and further 60% of the biomass energy partitioned into harvested seed as suggested by Zhu, Long, and Ort (2010) and supported by Koester et al. (2014). Therefore, there is a need to test the effect of supplemental N fertilizer in Iowa while using modern soybean cultivars and current management practices with the aim to boost soybean productivity. Most soybean studies in Iowa have focused only on final yield while paying little attention to the yield components that are important to understand the drivers of yield response to N fertilization.

Managing N in soybean production is a big challenge because of the difficulty in determining the temporal fluctuations between soil N mineralization and BNF during crop growth. However, a recent study on soybean BNF dynamics in Iowa (highly fertile soils and under rainfed systems) took place in nearby fields to the present experiment, and it was found that the contribution of BNF in Iowa soybean ranges from 23 to 65% of the total aboveground N demand (Córdova et al., 2019). Supplemental N fertilization seems to be a good option to decrease N limitations if these N supplies fall short (i.e., soil mineral N and BNF), providing more favorable production conditions. However, temporal synchronization is required for efficient use and effective application of N fertilizer in soybean. The timing of N fertilizer application is critical to benefit soybean production and be cost-effective, resulting in maximum economic return. Many studies have found that soybean accumulates 60% of total aboveground biomass N around the pod and seed development period (Bender et al., 2015; Gaspar, Laboski, Naeve, & Conley,

2017). Therefore, it is crucial to know and understand soybean phenology and BNF, particularly, that the highest N demand of soybean occurs at the seed-filling period (Bender et al., 2015; Garcia & Hanway 1976; Sinclair & de Wit, 1975), and that BNF supplement N to plants almost 50% more than indigenous soil N up to the R2.5 stage and then decreases (Córdova et al., 2019; Harper, 1974). It has been found that maximum soybean BNF occurs around early seed-filling period, but soon after decreases (Córdova et al., 2019; Deibert, Bijeriego, & Olson, 1979; Zapata, Danso, Hardarson, & Fried, 1987), thus portraying an N limiting scenario. Afza, Hardarson, Zapata, and Danso (1987) conducted a study on N limitation in soybean by quantifying the effect of N fertilizer at planting and at pod-filling stage with the use of unlabeled and labeled ^{15}N fertilizer. They found that when adding a starter N fertilizer at planting (20 kg N fertilizer ha^{-1}) BNF in soybean was not affected when applying additional N fertilizer at lower rates (e.g., 40 kg N ha^{-1}) at a later stage such as during pod setting, but higher rates did (e.g., 80 kg N ha^{-1}), which also reduced N uptake efficiency by 12%. We can speculate that soybean receiving N fertilizer at or after the pod-setting period could extenuate the N limitation acting as a complementary N supply while BNF decreases naturally, while at the same time it boosts pod-filling instead of vegetative growth controlling possible lodging (Afza et al., 1987; Wallace, Blanchet, Bouniols, & Gelfi, 1990).

Moreover, an ideal supplemental N fertilizer rate should match the amount of N exported in the seed at harvest, which is 80 kg N Mg^{-1} (Salvagiotti et al., 2008), with a maximum uptake rate raging between 4 and 6 kg N $\text{ha}^{-1} \text{d}^{-1}$ (Córdova et al., 2019). However, while determining optimal N fertilizer amendment doses to soybean we need to acknowledge potential N losses (e.g., leaching, volatilization, and runoff) due to different factors such as application form, timing, soil temperature, and moisture conditions when applying, etc. (Jaynes, Colvin, Karlen, Cambardella, & Meek, 2001; King, Williams, & Fausey, 2016; Parkin & Kaspar, 2006; Robertson, 2000).

It is reasonable to hypothesize that N fertilizer application during pod setting would increase soybean yield and yield components compared with not applying N (Wesley et al., 1999). Assuming a response, supplemental N fertilizer cost could be offset by higher soybean yield. This study aims to provide new information on soybean production and N partitioning in Iowa when using a modern soybean cultivar and supplemental N fertilizer at different times and rates. A field experiment was conducted to (a) analyze the effect of supplemental N fertilization additions on soybean yield and yield components, (b) evaluate the effect of supplemental N fertilization derived from different N rates

and timing of application on soybean dry matter and N partitioning, and (c) calculate the economic net return to N fertilizer application.

2 | MATERIALS AND METHODS

2.1 | Experimental sites

A field experiment was conducted in 2015 and 2016 at the Iowa State University Research Farms in Boone County, Iowa (2015, 42°0'1" N, 93°7'3" W; 2016, 42°0'38" N, 93°44'1" W). Sites were in maize–soybean rotation, with rainfed maize grown the previous year of the experiment. The soil for both sites is predominantly a Canisteo clay loam (fine-loamy, mixed, super active, calcareous, mesic Typic Endoaquolls according to the USDA Soil Taxonomy, USDA-NRCS, 2016). Soil tests pre-planting for both sites at the 0- to 30-cm depth averaged pH 7.2, 3.8% soil organic matter, and 15.3 mg kg^{-1} $\text{NO}_3\text{-N}$. Temperature and growing season rainfall dynamics at both sites are summarized in Figure 1.

2.2 | Experimental design and treatment application

The experimental design was a randomized complete block. Treatments included two factors: N fertilizer rate and N application timing. Treatments were randomly assigned within blocks that were replicated four times, with the inclusion of control plots (one per block; No-N). The fertilizer N source was ammonium nitrate (34% N), which was broadcast on each plot at one of three rates: 45, 90, and 135 kg N ha^{-1} . The application of N fertilizer was made at three different times: planting, flowering (R1/R2), and pod setting (R3/R4) (Fehr, Caviness, Burmood, & Pennington, 1971). Application timings were chosen stages prior to the highest soybean N demand at pod-filling (i.e., R4/5 stage) to complement any decrease of BNF around that time (Córdova et al., 2019; Garcia & Hanway, 1976) and the N application at planting served as a contrast. Although we aim to test different rates across a range to identify if there is an optimal rate when benefits are maximized. The individual plot size was 6.10-m width by 15.24-m length with a 0.76-m row spacing. Field plot layout and plot treatments remained the same in both years. Weed control and cultural practices were typical for soybean at each site. The soybean cultivar was Pioneer P92Y75 (maturity group 2.7), with a target seeding rate of 346,000 seeds ha^{-1} . Crop planting, herbicide application, plant sampling, and harvest dates are reported in Table 2.

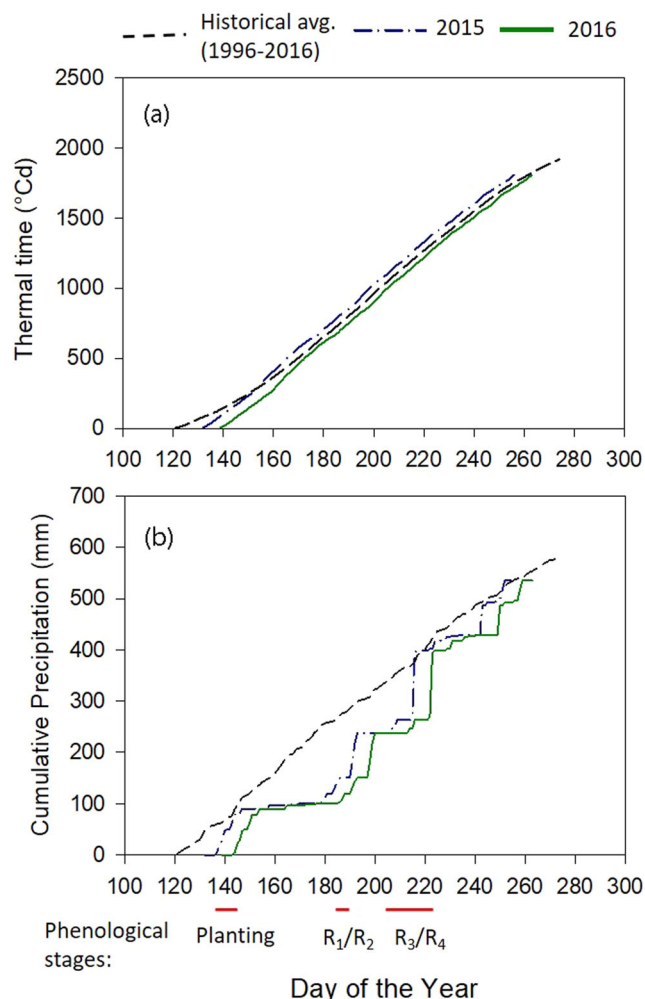


FIGURE 1 Thermal time (a) and cumulative precipitation (b) throughout the growing season 2015 (solid blue line), 2016 (solid green line), and historical average 1996–2016 (dotted black line; Data source: Mesonet, ISU, 2017). Thermal time calculated as $GDD = 0.5 \times (T_{max} + T_{min}) - T_b$, in which T_b is equal to 8°C

2.3 | Soybean biomass sampling and chemical analyses

Soybean plants were sampled for aboveground biomass at physiological maturity (i.e., R7/R8 stage, Fehr et al., 1971). The area sampled from each plot was 1 m^2 randomly selected from the second row to determine total aboveground biomass and plant counts (hereafter referred to as “hand-measured biomass”). A parallel study on root sampling conducted in nearby locations by Ordóñez et al. (2018) found that soybean root mass and length decline exponentially, on average 72% as distance from crop row increases; particularly beyond 0–0.10 m from the row. Thus, a crossover effect from neighboring treatments might have been negligible due to the distance (i.e., 1.52 m far from the neighboring plot). Hand-harvested

biomass was cut at ground level (Gan, Stulen, Posthumus, van Keulen, & Kuiper, 2002). Soon after, fresh weight of all hand-measured biomass samples was recorded. A subsample of 10 plants was taken from each plot, partitioned into seeds, pods, and stems (including petioles); oven-dried at 60°C until 0% moisture. Sample dry weight is reported per-area basis (g d.w. m^{-2}). No leaf weights were collected due to hand-measured biomass after leaf senescence. The number of pods m^{-2} , seeds m^{-2} , and individual seed dry weight were measured. Final soybean yield was reported using two methods: (a) hand-measured yield from 1 m^2 as described above, and (b) machine-measured yield obtained by harvesting the four middle rows of each plot using a combine with a Harvest Master (Juniper Systems) weigh system (between 21 and 26 d after the hand-measured sampling; Table 2). Both yields were reported in kg m^{-2} adjusted to 130 g kg^{-1} moisture. All plant organs were finely ground and subsampled for total carbon (TC) and total nitrogen (TN) concentrations determined by dry combustion (LECO CHN-200 Analyzer, LECO Corp.). All plant organ N concentration was reported as g N kg^{-1} dry weight, whereas organ N accumulation was reported on an area basis by multiplying the N concentrations and dry weight from each organ, then divided by the number of plants m^{-2} hand-harvested. Aboveground dry weight and N accumulation, reported in g m^{-2} , were calculated as the sum of all plant organ dry weights and N accumulated, respectively. Nitrogen harvest index (NHI) was calculated by dividing seed N accumulation by the total aboveground biomass N accumulation reported on an area basis from hand-measured samples.

2.4 | Calculations and data analysis

Descriptive statistics were used to summarize results. An analysis of variance was done by using Proc GLIMMIX procedures in SAS version 9.4 (SAS Institute), for each year separately because the experiment was set up in different plots from adjacent fields each year. First, the statistical analysis was done by excluding control plots (No-N treatment), and considering block as a random effect, in this way we study solely the effect of N fertilizer at different timings. Least squared means were tested for the one- and two-way interaction (N rate $\times$ application time), sliced by rate and time. All statistical tests were performed at the $\alpha = .05$ level of significance. Second, the No-N treatment was compared with the N-fertilized treatment (N rate $\times$ timing) mean with a single degree of freedom contrast. First-order linear regression was used to determine the measured soybean yield and yield component response to different N rates $\times$ year using PROC REG in SAS version 9.4 (SAS Institute), in which observations were the

TABLE 2 Effect of N fertilizer application on soybean yield and yield components. Data represent averages ($n = 4$) of each N treatment (application timing $\times$ rate) measured by year at the end of the season

		Individual seed			Yield ^a	
Application timing	N rate	seed	No. seeds	No. pods	Hand-measured	Machine-measured
	kg N ha ⁻¹	g seed ⁻¹	m ⁻²		kg m ⁻²	
2015						
Control	0	0.17	1,960.8	882.3	0.39	0.43
At planting	45	0.17	1,984.0 B ^b	919.8	0.38 B	0.45
	90	0.17	2,330.8 B	1,049.0	0.45 AB	0.42
	135	0.17	2,978.5 A	1,305.4	0.55 A	0.39
At R1/R2	45	0.16	1,995.7 B	1,011.0	0.36 B	0.43
	90	0.19	1,844.9 B	1,042.6	0.40 AB	0.42
	135	0.17	2,463.0 A	982.6	0.47 A	0.44
At R3/R4	45	0.18	1,742.3 B	921.9	0.37 B	0.41
	90	0.18	1,943.7 B	1,088.5	0.40 AB	0.43
	135	0.16	2,148.5 A	950.5	0.39 A	0.43
Comparison	Factor	<i>P</i> ≤ .05				
All Plus-N treatments	N rate (N)	ns	.01	ns	.03	ns
	Timing (T)	ns	ns	ns	ns	ns
	N × T	ns	ns	ns	ns	ns
Plus-N vs. No-N		ns	ns	ns	ns	ns
2016						
Control	0	0.17	2,173.3 B	1,133.3	0.41 B	0.40
At planting	45	0.16 b	2,634.3 A	1,257.1 abc	0.47 A	0.38
	90	0.15 b	3,005.0 A	1,178.7 bc	0.51 A	0.39
	135	0.19 a	2,854.3 A	1,396.5 abc	0.61 A	0.39
At R1/R2	45	0.16 ab	2,520.5 A	1,199.2 abc	0.47 A	0.37
	90	0.17 ab	2,500.5 A	1,305.8 abc	0.48 A	0.39
	135	0.16 b	2,948.9 A	1,537.6 a	0.53 A	0.39
At R3/R4	45	0.17 ab	2,704.7 A	1,410.5 ab	0.53 A	0.38
	90	0.15 b	3,190.5 A	1,442.7 ab	0.53 A	0.40
	135	0.16 ab	2,387.1 A	1,063.9 c	0.44 A	0.37
Comparison	Factor	<i>P</i> ≤ .05				
All Plus-N treatments	N rate (N)	ns	ns	ns	ns	ns
	Timing (T)	ns	ns	ns	ns	ns
	N × T	<.001	ns	.04	ns	ns
Plus-N vs. No-N		ns	.03	ns	.04	ns

Note. ns, not statistically significant.

^a Soybean yield was reported at 130 g kg⁻¹ moisture.

^b Different lowercase letters indicate N $\times$ T effect; whereas uppercase letters indicate significant differences from a single factor effect across N fertilizer treatments averages. A separate statistical comparison was done between Plus N vs. No-N treatments, and uppercase letters indicate differences between overall means within each year ($p < .05$).

means of each variable measured in each of the N rates across application timing. Furthermore, a *t*-test was used to determine whether linear regression slopes were significantly different among years with the use of the function CONTRAST and test equal slopes within PROC GLM (Supplemental Table S2). Thermal time (i.e., cumulative grow-

ing degree days) was calculated as follows:

$$\text{Thermal time} = 0.5 \times (T_{\max} + T_{\min}) - T_b \quad (1)$$

where $T_{\max}$ and $T_{\min}$ correspond to the maximum and minimum temperature, respectively (°C), and T_b is the

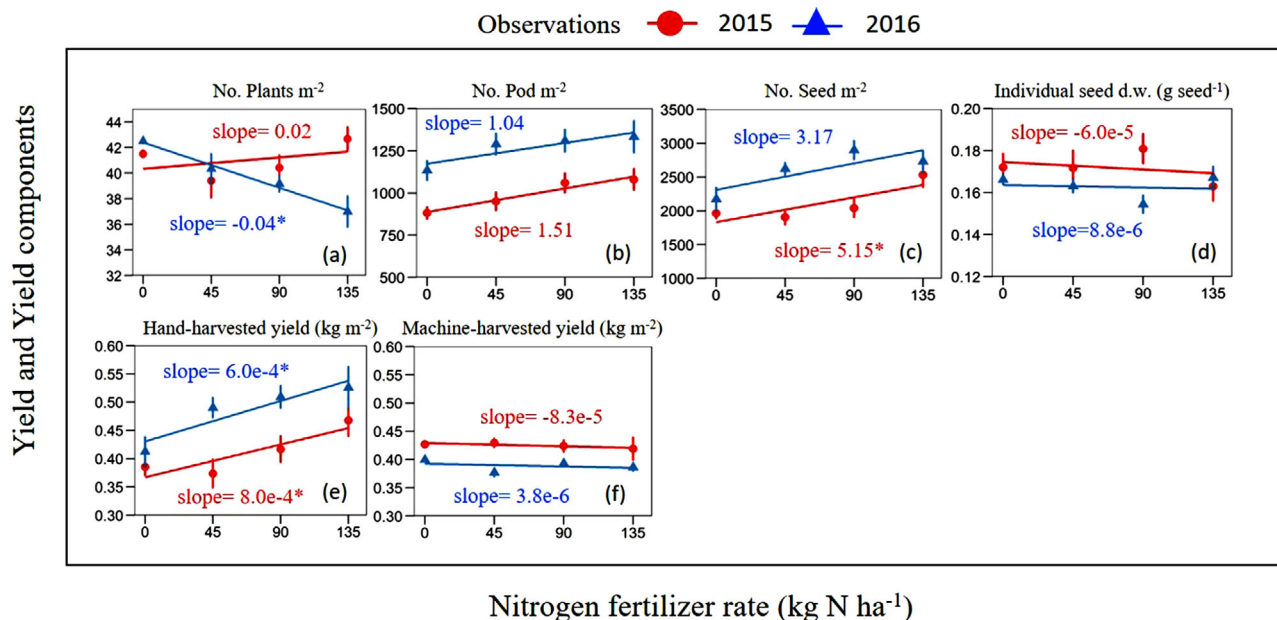


FIGURE 2 Soybean yield and yield components calculated per m^2 . Data are means averaged across the different application times ($n = 12$ for all N rates except control, $n = 4$, standard error bar) from 2015 (red circles), and 2016 (blue triangles). Solid lines are the best linear fit for the N fertilizer rates of each year. Asterisks indicate the significance of the slope coefficient ($p < 0.05$)

base temperature for soybean equal to 8°C (Wu et al., 2015). Cumulative precipitation during 2015 and 2016 were compared with historical precipitation date from 1996 to 2016, similarly as for thermal time (Figure 1). A simple economic analysis of return to N fertilizer cost was calculated as the net return above fertilizer cost similar to Salvagiotti et al. (2008), and by solely using the hand-measured yields reported in kg m^{-2} and at 130 g kg^{-1} moisture.

$$\begin{aligned} \text{Economic net return} &= (\Delta \text{yield} \times \text{market price}) \\ &\quad - (\text{additional N fertilizer} \\ &\quad \times \text{N price}) \end{aligned} \quad (2)$$

The market price of soybean in the United States during 2015–2016 were $\$0.33$ and $\$0.34 \text{ kg}^{-1}$ seed, and the price of N fertilizer was $\$1.04$ and $\$0.88 \text{ kg}^{-1}$ N fertilizer as ammonium nitrate, respectively (Iowa State University, 2018).

3 | RESULTS

3.1 | Seed yield and yield components

During 2015, the addition of N fertilizer affected positively the number of seeds m^{-2} , and hand-measured yield (Table 2). The N fertilizer application of 135 kg N ha^{-1} statistically increased both yield components disregarding timing of application. Number of seeds m^{-2} in 2015 was

positively correlated with N fertilizer rates (slope = 5.15, Figure 2c) in which the addition of $135 \text{ kg N fertilizer ha}^{-1}$ resulted in 28% higher number of seed m^{-2} compared with the other treatments (on average 2,530 vs. 1,974 seeds m^{-2} , respectively). Additionally, hand-measured yields were also positively correlated with N fertilizer rates during both years 2015 and 2016 with similar slopes (slope = $8.0\text{e-}4$ and $6.0\text{e-}4$, respectively, Figure 2e) in which during 2015 with the addition of $135 \text{ kg N fertilizer ha}^{-1}$ resulted in a 19% higher yield compared with the other N rates (on average 0.47 vs. 0.39 kg m^{-2} soybean yield reported at 130 g kg^{-1} moisture, respectively; Table 2). However, during both years, the machine-measured yield was not affected by N fertilizer rate, application time, nor their interaction (Table 2, Figure 2f). Plant stand density during both years was not affected by N fertilizer rate, application timing, nor their interaction (Table 2). However, during 2016 the plant stand density across all N fertilized treatments was lower than the No-N (approx. average 39 vs. 43 plants m^{-2} , respectively, Table 2), with a decreasing rate of -0.04 plants m^{-2} per kg N fertilizer (Figure 2a).

In 2016 the interaction of N fertilizer addition and application timing increased individual seed dry weight and the number of pods m^{-2} (Table 2). Both were 17 and 20% higher when applying 135 kg N ha^{-1} at planting and R1/R2 stage, respectively (on average 0.19 g seed^{-1} , and $1,537.6$ pods m^{-2} ; Figure 2d and 2b). However, when applying 45 kg N ha^{-1} at R3/R4 stage, results were only 8% lower

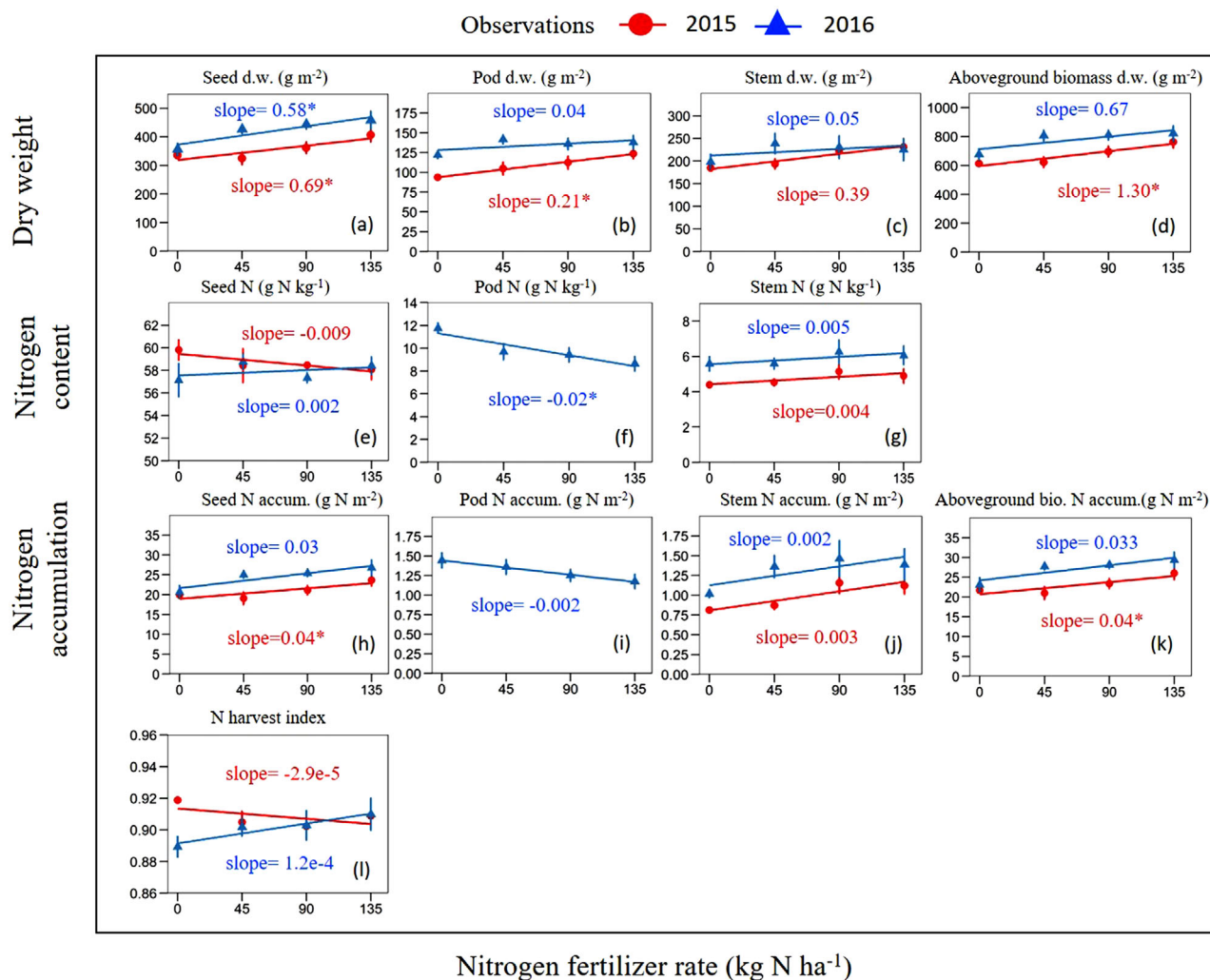


FIGURE 3 Soybean dry weights, N content, and N accumulation reported from hand-measurement per m². Data are means averaged across the different application times ($n = 12$ for all N rates $\pm$ standard error) from 2015 (red circles), and 2016 (blue triangles). Solid lines are the best linear fit for the N fertilizer rates of each year. Asterisks indicate the significance of the slope coefficient ($p < 0.05$)

than the ones obtained with 135 kg N ha⁻¹ at their respective timing (on average 0.17 g seed⁻¹ and 1,410.5 pods m⁻²; Table 2).

Soybean yield collected from a small area (i.e., hand-measured at 1 m²) and yield reported by machine harvesting from a large area (93 m²) did not show the same response to N fertilization. This is not surprising given the size of the harvest area and the difference in harvest time. The yield variability was high for the hand-measured area and much smaller for the machine-measured area. Additionally, the hand-measured occurred approximately 21–26 d before the mechanical harvest. During that period, rainfall occurred twice in 2015 and seven times in 2016, contributing 14 and 84 mm cumulative precipitation. Moreover, plant lodging was observed in the field that likely resulted in yield loss by the machine-measured yield.

3.2 | Soybean seed weight, nitrogen concentration, and accumulation in response to nitrogen fertilizer

Results showed that only the application of supplemental N fertilizer in soybean increased seed and aboveground biomass dry weight during both years (Table 3). Both years seed dry weight had a linear response to N fertilizer rates (Figure 3a). During 2015 the application of supplemental N fertilizer of 135 kg N ha⁻¹, regardless of application timing, increased seed and aboveground biomass dry weight by 19 and 16%, respectively, higher than the results from the other N rates (on average 406.4 g seeds d.w. m⁻², and 761.3 g aboveground biomass d.w. m⁻²; Table 3). In 2016, aboveground biomass dry weight was high also when applying 135 kg N ha⁻¹ but only when it was applied at planting (on average 917.8 g m⁻²), and only 3% less than that

TABLE 3 Effect of N fertilizer application on soybean biomass dry weight, N concentration and N accumulation in aboveground plant organs, and yield. Data represent averages ($n = 4$) of each N treatment (application timing $\times$ rate) measured by year at the end of the season

Application timing	N rate kg N ha ⁻¹	No. Plants no. m ⁻²	Dry weight				N content				N accumulation			
			Total			Seeds	Pods	Stems	Seeds	Pods	Stems	Total		
			Seeds	Pods	Stems								aboveground	aboveground
2015														
Control	0	41.5	335.1	93.6	184.5	613.2	59.8	–	4.4	19.9	–	0.8	21.7	0.9
At planting	45	38.8	332.1 B ^a	105.6	188.4	626.1 B	56.3	–	4.4	18.6	–	0.8	20.5	0.9
	90	40.0	393.2 AB	113.4	238.4	745.1 AB	58.4	–	5.3	23.0	–	1.3	25.4	0.9
At R1/R2	135	43.8	475.2 A	142.8	262.0	880.0 A	58.2	–	4.4	27.7	–	1.1	30.2	0.9
	45	41.5	316.7 B	97.5	201.7	616.0 B	61.4	–	4.9	19.8	–	1.0	21.8	0.9
	90	40.3	344.7 AB	115.5	197.6	657.8 AB	59.1	–	4.8	20.3	–	0.9	22.4	0.9
At R3/R4	135	43.5	406.9 A	122.9	226.9	756.7 A	58.5	–	5.7	23.8	–	1.3	26.3	0.9
	45	38.0	323.8 B	111.2	190.0	625.0 B	57.6	–	4.3	18.7	–	0.8	20.6	0.9
	90	41.0	346.3 AB	107.7	229.0	683.0 AB	57.9	–	5.4	20.0	–	1.3	22.4	0.9
Comparison	135	40.8	337.2 A	105.1	205.1	647.3 A	57.6	–	4.6	19.4	–	0.9	21.4	0.9
	Factor	$P \leq .05$												
All Plus-N treatments	N rate (N)	ns	.03	ns	ns	.04	ns	–	ns	ns	–	ns	ns	ns
	Timing (T)	ns	ns	ns	ns	ns	ns	–	ns	ns	–	ns	ns	ns
Plus-N vs. No-N	N × T	ns	ns	ns	ns	ns	ns	–	ns	ns	–	ns	ns	ns
		ns	ns	ns	ns	ns	ns	–	ns	ns	–	ns	ns	ns

(Continues)

TABLE 3 (Continued)

Application timing	N rate	No. Plants	Dry weight				N content				N accumulation					
			Seeds	Pods	Stems	Total aboveground	Seeds	Pods	Stems		Seeds	Pods	Stems	Total aboveground	NHI	
2016																
Control	0	42.5	356.4 B	122.0	198.0	676.3	57.1	11.8 A	5.6		20.7	1.4	1.0	23.1		0.9
At planting	45	41.0	411.5 A	135.1	195.5	742.1 bc	60.0	9.9 B	5.3		24.6 ab	1.3	1.1	27.0 ab		0.9
	90	38.0	448.0 A	118.2	218.3	784.5 abc	57.5	10.5 B	7.8		25.7 ab	1.2	1.8	28.7 ab		0.9
At R1/R2	135	38.3	530.0 A	136.5	251.3	917.8 a	60.3	9.5 B	5.5		32.0 a	1.3	1.4	34.7 a		0.9
	45	40.3	407.2 A	132.5	250.1	789.8 abc	58.2	9.8 B	5.5		23.7 ab	1.3	1.4	26.4 ab		0.9
	90	40.3	421.6 A	134.4	217.4	773.4 abc	57.7	9.5 B	5.8		24.3 ab	1.3	1.3	26.9 ab		0.9
At R3/R4	135	36.5	462.9 A	156.2	265.3	884.4 ab	58.0	9.5 B	6.3		26.6 ab	1.4	1.8	29.8 ab		0.9
	45	39.8	460.8 A	157.2	270.1	888.1 ab	58.1	9.4 B	5.9		26.8 ab	1.5	1.6	29.9 ab		0.9
	90	39.3	461.9 A	154.8	255.8	872.6 ab	56.8	8.2 B	5.2		26.2 ab	1.3	1.3	28.8 ab		0.9
	135	36.3	381.1 A	120.3	159.0	660.3 c	56.9	7.0 B	6.4		21.7 b	0.8	1.0	23.6 b		0.9
Comparison	Factor	$P \leq .05$														
All Plus-N treatments	N rate (N)	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns
	Timing (T)	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns
Plus-N vs. No-N	N × T	ns	ns	ns	ns	.03	ns	ns	ns	ns	.05	ns	ns	.02	ns	ns
		ns	.04	ns	ns	ns	ns	ns	.04	ns	ns	ns	ns	ns	ns	ns

Note. NHI, nitrogen harvest index; ns, not statistically significant.

" Different lowercase letters indicate N × T effect; whereas uppercase letters indicate significant differences from a single factor effect across N fertilizer treatments averages. A separate statistical comparison was done between N vs. No-N treatments, and uppercase letters indicate differences between overall means within each year ($p < .05$).

when applying 45 kg N ha⁻¹ at R3/R4 stage (Table 1, Figure 3a, 3d). Moreover, aboveground biomass dry weight response to N fertilizer rates had similar slopes during both years (slopes = -0.69 and 0.58 g seeds d.w. m⁻² kg⁻¹ N fertilizer during 2015 and 2016, respectively; Supplemental Table S2). Even though there were not significant differences for other plant organ dry weights such as pod and stem, in 2015 pod dry weight was positively correlated with N rates (slope 0.21 g pod d.w. m⁻² kg⁻¹ N fertilizer; Figure 3b, Supplemental Table S2).

When comparing N fertilized plots with No-N plots, seed dry weight, and pod N concentration were significantly different in 2016 (Table 3); however, they had contrasting results. For instance, N fertilizer addition increased seed dry weight by 24% compared with the No-N treatment (on average 442.8 vs. 356.4 g seed d.w. m⁻², respectively, Table 3). Pod N concentration among all the organs was the only plant organ affected negatively by the addition of N fertilizer compared with the No-N plots (slope = -0.02 g N kg⁻¹ pod kg⁻¹ N fertilizer, Figure 3f), decreasing 21% of its N concentration (on average 9.25 vs. 11.76 g N kg⁻¹ pod; respectively; Table 3).

Nitrogen accumulation in seed and aboveground biomass benefited from the interaction of N fertilizer and timing during both years (Table 3). During 2015, seed and aboveground biomass N accumulation was positively correlated with N fertilization rates resulting with a N accumulation rate of 0.04 g seed N accumulation m⁻² and 0.04 g aboveground biomass N accumulation m⁻² per kg N fertilizer applied (Figure 3h and 3k, respectively). During 2016, the interaction between N fertilizer rate and application timing had a positive effect on both variables ($p = .048$ and $.023$, respectively; Table 3). Higher accumulation of N in seed and aboveground biomass was obtained by applying 135 kg N ha⁻¹ at planting (on average 32.0 and 34.7 g N uptake m⁻², respectively), being both 1.5-times higher than No-N plots (Table 3). However, when applying 45 kg N ha⁻¹ at R3/R4 stage during 2016, both seed and aboveground biomass N accumulation also increased N accumulation with a difference of 16 and 14% less than if applying 135 kg N ha⁻¹ at planting, respectively (Table 3), but still significantly higher than results from the other Plus-N and No-N treatments. Pod-, stem-N-accumulation, and nitrogen harvest index (NHI) were not affected by N fertilizer additions, application timing, nor their interaction in both years (Table 3, Figure 3i, 3j, and 3l).

3.3 | Economic net return to the nitrogen fertilizer applied to soybean

The economic net return to N fertilizer applied to soybean during both years was not statistically different for

any of the factors tested, such as N fertilizer rate, application timing, nor their interaction (p value > 0.05; Table 4). However, in both years all N fertilizer additions, regardless of application timing, produced on average 0.07 ± 0.03 kg m⁻² more soybean grain than No-N plots, with the exception of 45 kg N ha⁻¹ N fertilizer during 2015 soybean yield, which was on average -0.01 kg m⁻² lower than No-N treatment (Table 4), resulting in the only treatment with a negative return of \$87.64 ha⁻¹ to N fertilizer application. All N fertilizer treatments had a considerably greater economic gain that ranged from \$5.33 to \$281.89 ha⁻¹ (except the 45 kg N ha⁻¹ treatment during 2015, Table 4).

4 | DISCUSSION

This study brings detailed data on N fertilization effects on soybean yield that updates the scientific literature while using a modern soybean cultivar. The last research on N fertilization in Iowa was a decade ago (Ruiz Diaz et al., 2009). Significant increases in seed yield and yields components (Table 2 and Figure 2), as well as soybean biomass dry weight, N concentration, and N accumulation, were obtained as a response to N fertilizer additions (Table 3 and Figure 3). Moreover, the detailed seed component analysis added value in understanding the lack of response to N fertilizer applications when only looking at machine-measured yields (Table 2). These results were opposite to earlier findings on the topic, whereas the economic analysis illustrated the magnitude of revenue gained by applying N fertilizer on soybean (Table 4).

The yield response to N was variable from year to year as having been influenced by different seasonal growing conditions such as precipitation. For instance, 2016 was 1.5 °C warmer, and had 100 mm less rain than 2015 (Figure 1). Water-logged soil early in the growing season not only affected crop growth (Kucharik & Serbin, 2008; Stanford & Epstein, 1974) but also could have resulted in greater losses of N from the system (Blann, Anderson, Sands, & Vondracek, 2009). In 2016, more favorable weather (i.e., warmer and 100 mm less rain than 2015; Figure 1) preserved the availability of the N fertilizer applied, similar to findings by Osborne and Riedell (2006). Consequently, soybean seed weight was 20% higher in 2016 than in 2015 (Figure 2d) because of a higher number of pods and seeds m⁻² in 2016 (Figure 3b and 3c, respectively), direct consequences also found by Gaspar and Conley (2015). A growing season with adequate environmental conditions such as temperature, rainfall, inorganic N influence crop productivity (Egli & Hatfield, 2014). For instance, few studies found a negative correlation between soybean yield and increasing late-summer temperatures for the interval between late July and early August (Willhem &

TABLE 4 The economic return to N fertilizer applied to soybean at different rates by year. Soybean yield, yield difference, and potential net return to N cost are data means ($n = 12 \pm$ standard error)

Year	Expenses		Income			Potential net return to N cost ^b
	N rate	Price N ^a	Yield	Yield difference	Price seed	
	kg N ha ⁻¹	US\$ kg ⁻¹ N	kg m ⁻²	+N minus No-N	US\$ kg ⁻¹ ^a	US\$ ha ⁻¹ yield
2015						
	No-N	1.04	0.39 ± 0.03		0.33	
	45	1.04	0.37 ± 0.02	-0.01 ± 0.02	0.33	-87.64 ± 81.47
	90	1.04	0.42 ± 0.02	0.03 ± 0.02	0.33	5.33 ± 76.27
	135	1.04	0.47 ± 0.03	0.08 ± 0.03	0.33	127.73 ± 87.16
2016						
	No-N	0.88	0.41 ± 0.05		0.34	
	45	0.88	0.49 ± 0.02	0.08 ± 0.02	0.34	236.97 ± 60.77
	90	0.88	0.51 ± 0.02	0.10 ± 0.02	0.34	265.79 ± 68.04
	135	0.88	0.53 ± 0.04	0.12 ± 0.04	0.34	281.89 ± 123.87
						$P \leq .05$
			Comparison	Factor/Year	2015	2016
			All Plus-N treatments	N rate (N)	ns	ns
				Timing (T)	ns	ns
				N × T	ns	ns

Note. ns, not statistically significant.

^a Values obtained from Marketing Price: Iowa Cash Ag Decision ISU (Iowa State University, 2018).

^b Potential return to N cost = [(Yield difference × price seed) - ((N rate/10,000) × price N)] × 10,000.

Wortmann, 2004; Yamoah, Walters, Shapiro, Francis, & Hayes, 2000), which could have been around R4 stage and the beginning of R5 (Córdova et al., 2019), thus, significantly impacting soybean production during early stages prior to pod filling (Willhem & Wortmann, 2004). Moreover, optimal soil temperature for soybean BNF and nodule formation ranges between 20 and 25 °C (Lindemann & Ham, 1979; Purcell, Serraj, Sinclair, & De, 2004). Therefore, during the soybean growing season of 2016, perhaps crops achieved higher photosynthetic rates during seed-filling periods resulting in gains of soybean yield as it was found by De Bruin, Singer, Pedersen, and Rotundo (2010).

Our results agree with findings from other studies in which soybean yield variability is mostly governed by the environment that influence the positive effect of N fertilizer additions. For instance, Tamagno et al. (2017), in a synthesis of soybean studies and nutrient partitioning, found that 33 and 22% of the variation of soybean aboveground biomass and seed yield were attributed to the environment more than management. Subsequently, Mourtzinis et al. (2018) in a recent review study of N fertilizer trials in the United States, corroborate that most yield variability is derived from environmental factors (68%), and less than 1% attributed to supplemental N fertilizer addition. Perhaps,

all these conclusions were mostly based on results from machine-measured yield, which could be underestimating the actual gains in yield and underappreciate increases in other soybean yield components. Moreover, contrasting results obtained between hand-measured and machine-measured yields, caution should be exercised in interpreting yield response, not only for N fertilization studies but for any other experiments. In this study, the effect of the yield data collection method may influence grain yield production more than the N fertilizer applications. Unfortunately, we failed to record plant lodging and seed losses pre- and post-machine harvester. Additionally, we do not know about any study that have recorded yield losses at the field when sampling with a machine harvester.

There was a consistent positive effect of the N fertilizer application of 135 kg N ha⁻¹ overall measurements (i.e., whole aboveground biomass dry weight, N concentration, N accumulation, seed yield, and yield components); however, the application timing was not consistent from one year to the next. In 2015, N application affected all measurements, whereas in 2016, this N rate was effective when applied at planting in all measurements except in seed yield components in which the effect was positive regardless of the rate and the application timing (i.e.,

Plus-N > No-N; Table 2). However, during 2016 the application of 45 kg N ha⁻¹ at R3/R4 stage, closely followed results obtained when applying 135 kg N ha⁻¹ at planting, suggesting a more promising N treatment to be applied to soybean in the time of higher N demand (Gaspar et al., 2017; Wesley et al., 1999). Interestingly, another study conducted in Kansas during 2015–2016 by Ortíz et al. (2019) found that N fertilizer application at R3 stage of 56 kg N ha⁻¹ increased soybean yield by 9%, and when applying a large amount of 670 kg N ha⁻¹ in three applications of 223 kg N ha⁻¹ at three different times during the growing season (planting, R1, and R3 stages). These results reinforce the idea that a lower application of N fertilizer at or near pod setting stage can catch up when BNF decreases and meet the soybean N needs. However, there is a need to expand the investigation during this time of high N demand in soybean production. Application of a lower rate of N fertilizer at R3/R4 will not only increase yield but also will eliminate any trade-off when applied earlier in the season affecting nodulation and will decrease any significant negative impact on the environment.

Promising results were obtained from the potential economic net return to N fertilizer applied to soybean (Table 4). Market prices of soybean over the last decade have motivated farmers and agronomists to look for strategies that can boost yield, and the addition of N fertilizer is one option being considered. However, Orlowski et al. (2016) analyzed the relative change and break-even probabilities of several agronomic practices in nine states that produce soybean. Results showed that in Iowa (also Indiana and Illinois), the addition of N fertilizer leads to a relative yield change of 1.2% compared with standard practice. With high soybean prices $\geq \$0.55$ kg⁻¹ and yield levels ≥ 4 Mg ha⁻¹, there is a probability of break-even points between 2 and 6%. Looking beyond yield gains and economical gains, Greer et al. (2020) performed a comparative analysis of high-input (i.e., conventional, N fertilizer additions, higher seeding rate, pesticide application) vs. reduced-input and standard soybean management (lower the inputs of high-input, and No-N additions, respectively) in the United States during 2015–2017. Their results showed that yield gains in high-input management systems were not sustainable environmentally speaking (lower energy efficiency, increased the partial global warming potential, and agrochemical contamination risk) compared with the reduced-input and standard management. Moreover, previous studies have demonstrated that yield gains are site-specific due to specific environmental growing conditions (Rattalino Edreira et al., 2017), and economic gains depend mostly on seed price (Orlowski et al., 2016) when fossil fuel price has not changed.

5 | CONCLUSIONS

This study investigated the effect of supplemental N fertilizer on soybean yield and its components, which included different rates and timings across 2 yr (2015–2016) in central Iowa. During both years (2015–2016), soybean production responded to N fertilizer additions leading to produce higher dry weights of seed and total aboveground biomass, and higher N accumulation in seed, and total aboveground biomass, all reported on an area basis. Even though these positive effects of N fertilizer application resulted in soybean yield differences, it could not be captured by machine-measured yield. Results from this study show that soybean is responsive to N fertilizer applications as low as 45 kg N ha⁻¹ during pod set stage or 135 kg N ha⁻¹ at planting. However, the economic analysis showed that all N applications, except for 45 kg N ha⁻¹ during 2015, resulted in a positive net return to N fertilizer application to soybean. These results are specific findings that belong to our experimental conditions (i.e., one location in central Iowa and one genotype). This study contributes not only to update information of Iowa soybean related to the effect of N fertilizer additions over yield and yield components, but also reports on the N uptake partitioning while testing a modern soybean cultivar. The last N fertilizer trial in Iowa was reported almost a decade ago by Ruiz Diaz et al. (2009).

Confounding results obtained between hand-measured and machine-measured yields invite researchers to evaluate possible losses from machine-harvester to elucidate the underlying causes of yield differences between methods (e.g., quantify plant lodging and seed losses before and after the machine harvest). There is a clear need to improve biomass sampling strategy; perhaps start at R6 prior to leaf senescence and place a net underneath the sampling area to collect any biomass and seed loss while the plant is maturing.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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